

IBIT Overnight Strategy - INTRADAY ML Model

Generated: 2025-12-24 23:52

Key Advantage: This model uses SAME-DAY intraday data available at 3:50 PM, including today's gap, intraday return, and volume - providing real-time signal while still being usable for live trading.

ML Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	52.5%	54.1%	70.7%	0.61
Gradient Boosting	48.9%	51.6%	64.0%	0.57
Logistic Regression	53.9%	55.3%	69.3%	0.62
XGBoost	51.1%	53.6%	60.0%	0.57

Strategy Backtest Results (Test Period)

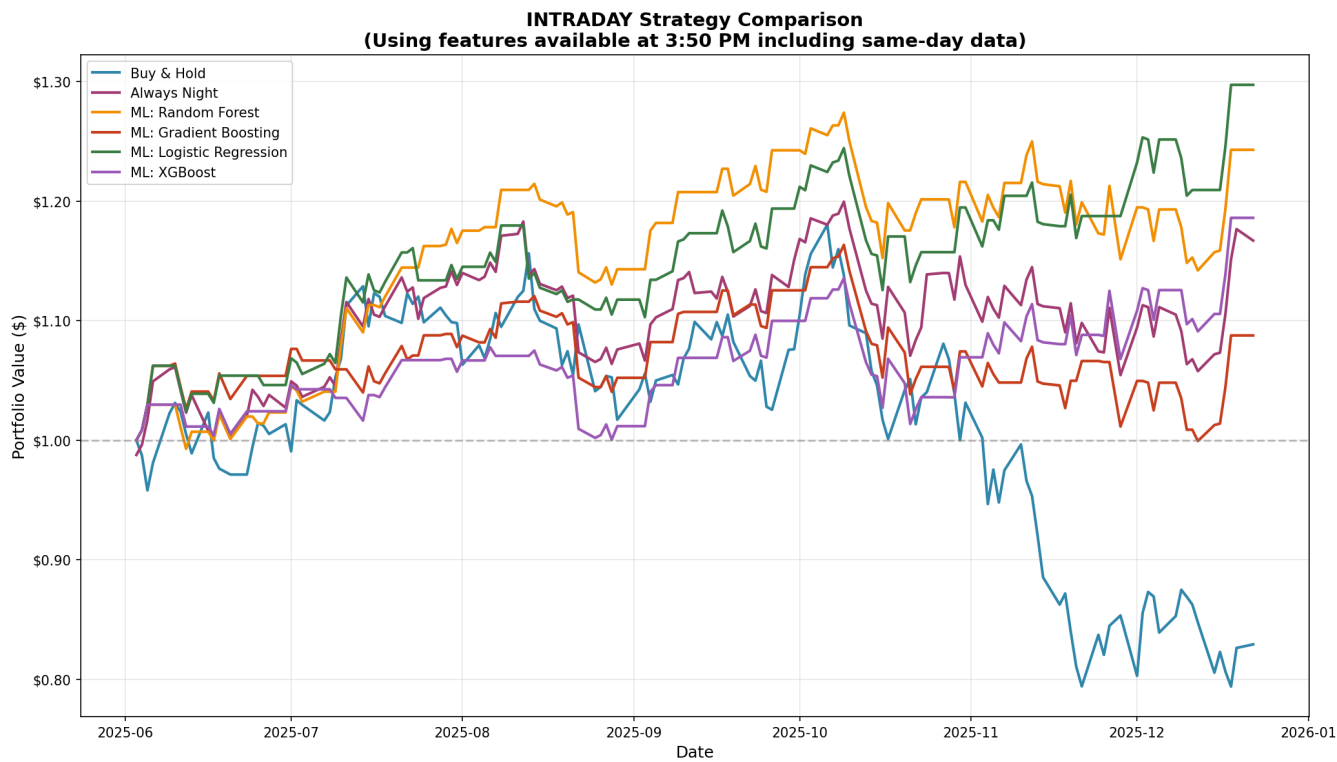
Strategy	Total Return	Sharpe	Max DD	Win Rate	Trades
Buy & Hold	-17.1%	-0.85	-32.7%	45.4%	141
Always Night	16.7%	0.92	-12.1%	53.2%	141
ML: Random Forest	24.3%	1.67	-10.4%	54.1%	98
ML: Gradient Boosting	8.8%	0.46	-14.1%	51.6%	93
ML: Logistic Regression	29.7%	2.26	-9.5%	55.3%	94
ML: XGBoost	18.6%	1.29	-10.7%	53.6%	84

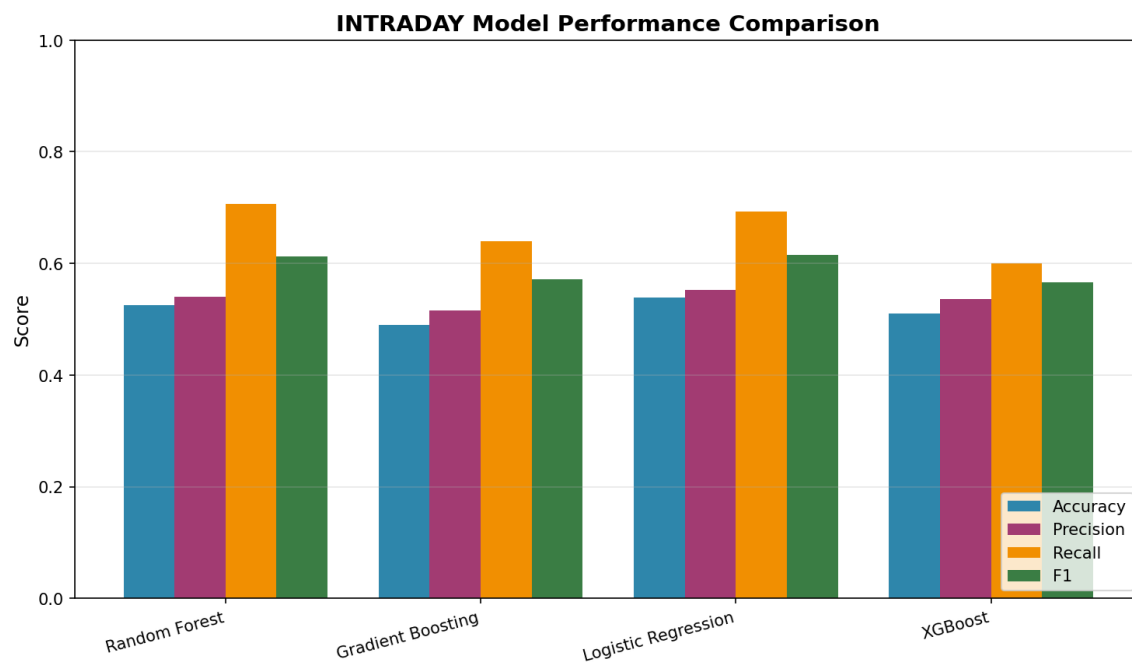
Key Intraday Features

Same-day features available at 3:50 PM:

- **today_gap**: $(\text{Today Open} - \text{Yesterday Close}) / \text{Yesterday Close}$
- **intraday_return**: $(\text{Current Price} - \text{Today Open}) / \text{Today Open}$
- **intraday_vs_spy**: IBIT intraday return minus SPY intraday return
- **intraday_range**: $(\text{Today High} - \text{Today Low}) / \text{Today Open}$
- **intraday_position**: Where current price is in today's range (0-1)
- **today_volume_vs_ma**: Today's volume vs recent average
- **current_vs_ma5/20**: Current price vs moving averages

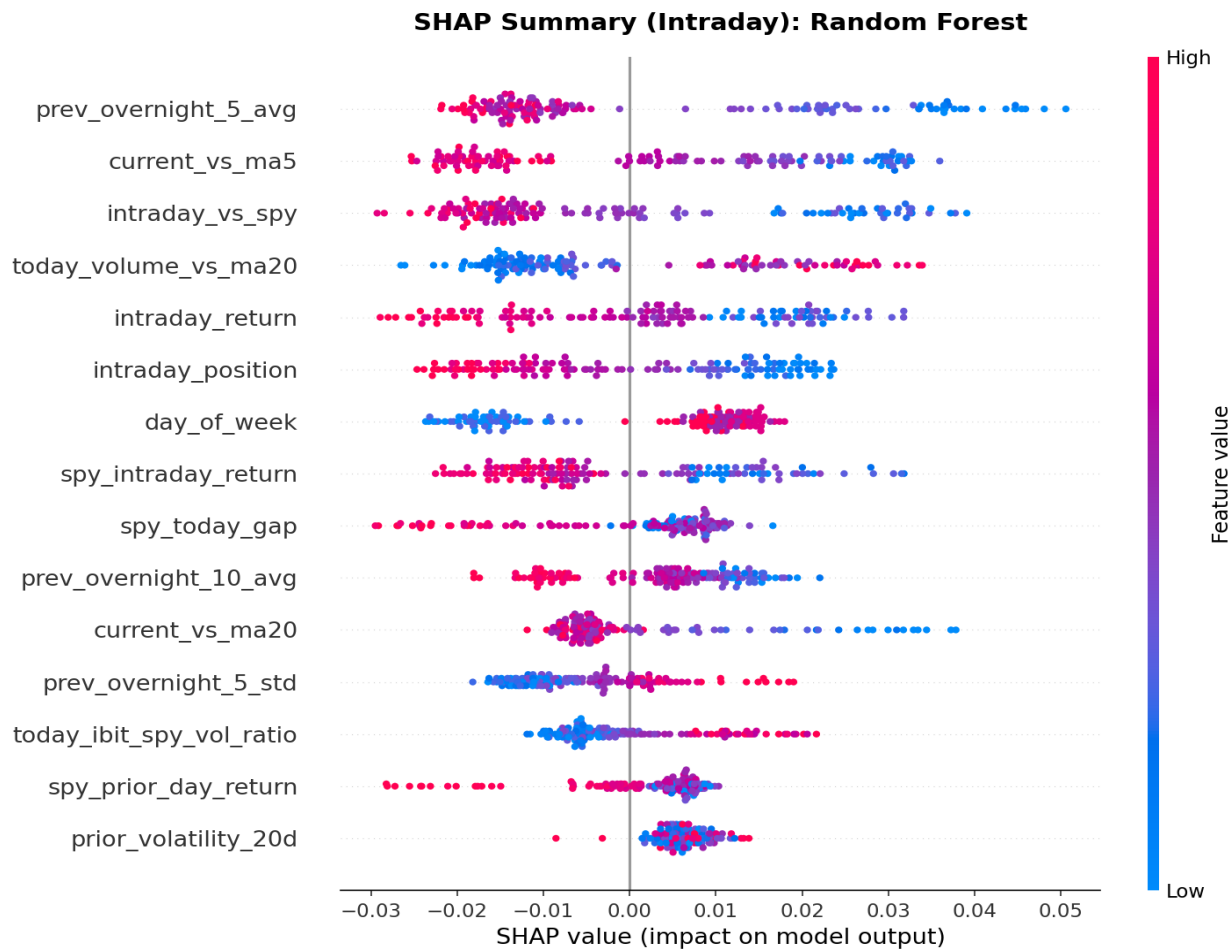
Performance Charts



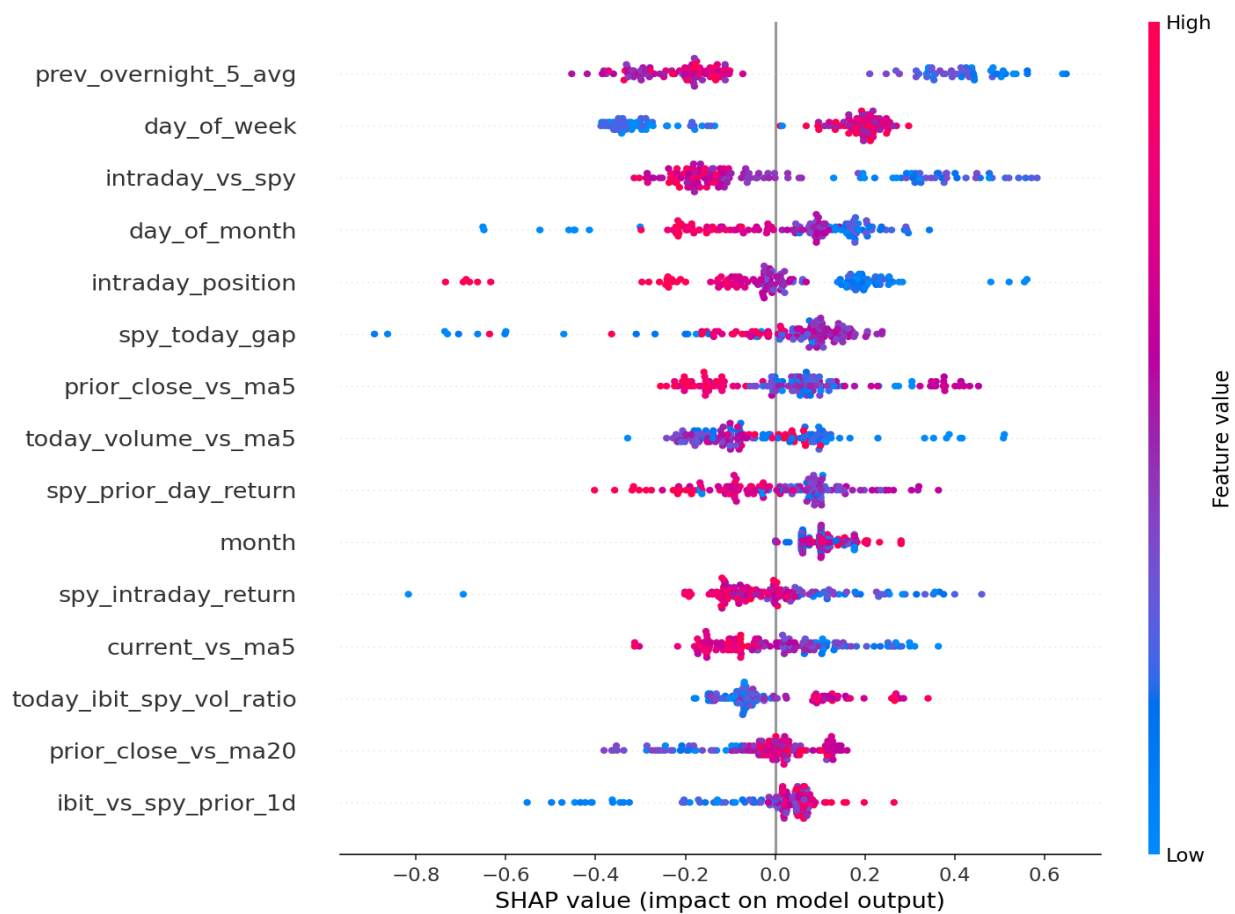


SHAP Analysis - Feature Importance

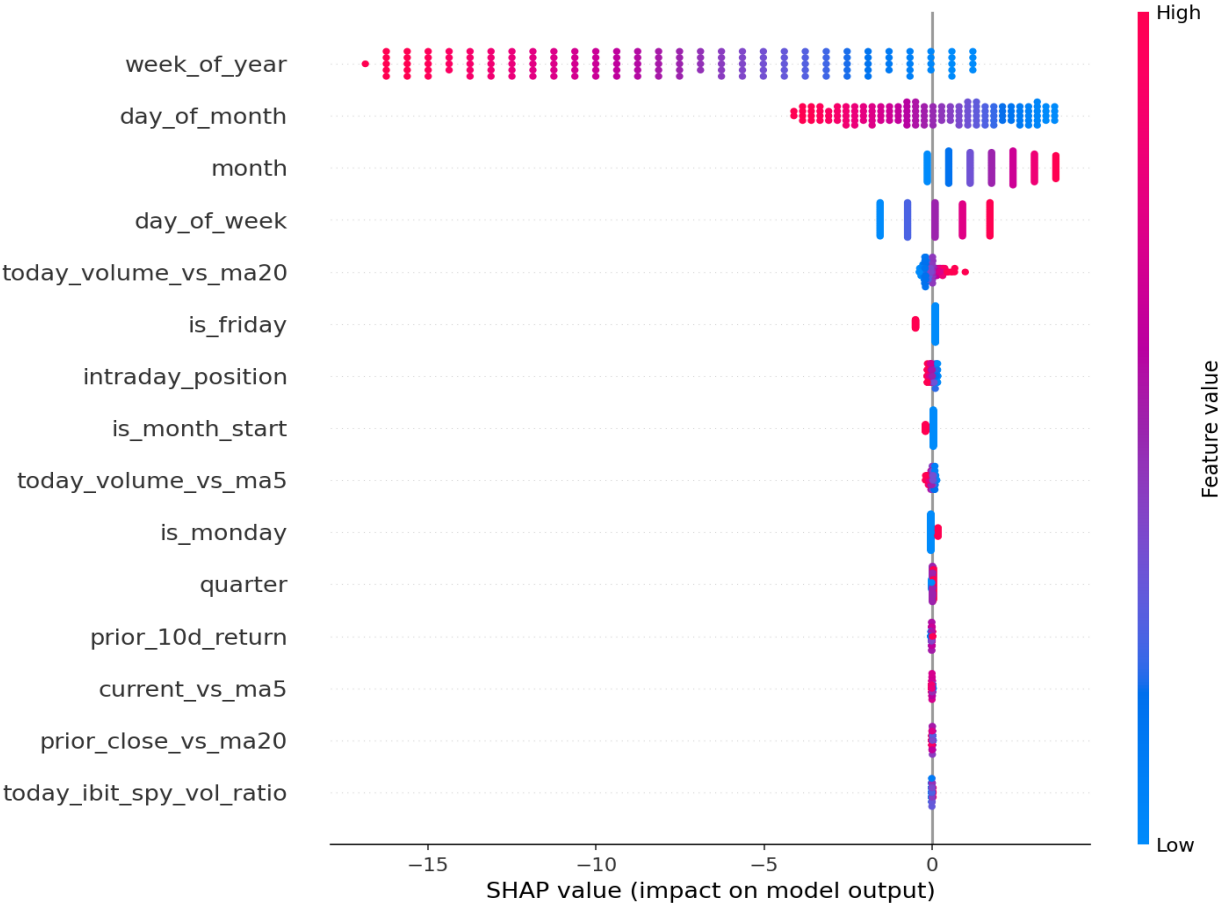
SHAP values show how each feature contributes to predictions. Look for same-day intraday features (gap, intraday_return, etc.) to see how much predictive power they add.



SHAP Summary (Intraday): Gradient Boosting



SHAP Summary (Intraday): Logistic Regression



SHAP Summary (Intraday): XGBoost

